

Mirror Placement Matters in Remote Collaboration

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Figure 1: Comparison of 3 different user interfaces for tele-conferencing. Left: Single View without a video feedback mirror, Middle: a tele-immersed video feedback view, Right: isolated video feedback mirror, 2022.

ABSTRACT

Video Feedback (VF) in tele-conferences is reported to contribute to ‘Zoom Fatigue’ (ZF). However, such feedback is important for many remote collaborative tasks where users must be aware of their relationship to the camera. Tele-immersion (TI) is presented as an alternative to traditional interfaces that can mitigate symptoms of ZF while maintaining the benefit of VF. The effects of TI on behaviour are understudied. Therefore, we present the findings of a novel field study of 14 domain experts performing a remote collaborative task—improvisational theatre—under all three conditions: 1) with video feedback (VF) in an isolated window, 2) within a tele-immersive environment, and 3) without video feedback at all. A qualitative study was conducted using surveys measuring improv performance metrics. ‘Physical engagement’ and ‘presence’ were perceived highest with tele-immersion, while ‘attunement’ and ‘flow’ between performers were comparable between tele-immersion and no mirror. Isolated VF was perceived worse for most conditions.

CCS CONCEPTS

• Human computer interaction ; • Interaction design; • Collaborative and social computing; • Visualisation;

KEYWORDS

tele-immersion, video feedback mirrors, tele-conferencing

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1 INTRODUCTION

There are non-trivial drawbacks to virtual meetings that continue to undermine user health, well-being, and enjoyment, many of which have been linked to the presence of video feedback (VF) mirrors. Recent studies on the phenomenon of ‘Zoom Fatigue’ [21] credit VF for increased negative self evaluation from staring at video of oneself [1, 24] and anxiety about appearance [1, 7, 12, 14, 20]. Others, however have demonstrated that removing VF from conferencing actually increased social anxiety [17]. Furthermore, VF plays an important role in many remote collaborative activities, including improvisational theatre where users must be aware of how they appear to each other on screen to inform a performance. For this purpose, we differentiate the traditional VF mirror of tele-conferencing—that allows one to see how they appear *to others*—from the *tele-immersive mirror* (TI), as defined by Galloway and Rabinowitz [6]—that allows one to see how they appear *with others*. We build on previous studies of Tele-Immersion which suggest TI might mitigate other factors related to Zoom Fatigue such as reduced mobility [1] and sense of social presence [22], where participants using TI mirrors reported feeling more physically active and

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‘there’ in a shared virtual space than in the ‘here’ of their physical environment [3].

This study contributes insights related to how 1) viewing oneself isolated from remote collaborators (typical mirror VF) compares with 2) viewing the self fully immersed in a virtual environment with a TI mirror or 3) not viewing the self at all. We evaluate these three conditions while participants engage in a collaborative task of improvisational theatre (an inherently *ludic* social activity [5]) in a custom-built virtual environment. We study the ludic response to video-conferencing layouts as it is an effective way to evaluate usability and engagement [2, 8, 18, 27]. It is the first study of its kind to comparatively measure mirror-based tele-immersion against traditional video conferencing layouts for their impact on the sense of *flow*¹, social *attunement*², physical engagement, sense of physical presence, and semantic choice between remote performers.

Specifically, we investigate whether tele-immersive ‘mirror’ configurations enhance: a) the sense of presence with the shared virtual environment, b) the experience of *flow*, c) the *attunement* between participants, d) the sense of playfulness and e) the improvisational choices of participants. Findings of this field study with fourteen improvisers indicate that the tele-immersive environment contributed to increased physical activity while performing, to a sense of physical connection with their partner, and to a sense of being transported into the same virtual space as their remote partner. Small differences were found in self-reports of flow or attunement between the three platforms; however, trends are identified for further research with larger sample sizes.

Semantic analysis of improvisational dialogue found a significant increase in the frequency of utterances referring to the ‘present’ moment between improvisers for tele-immersive environments over the other two conditions, as well as references to the given virtual environment of the scene. A qualitative analysis assisted with automated tools for text categorization found that interfaces affected the content and style of performance. When no mirrors were present, improvisers tended to speak in slightly longer sentences and included slightly more monologues on subjects related to imagined pasts or expected future events. When a small video mirror was present at the side, the scenes generally involved slightly more conflict with each other. Results are discussed to explain how tele-immersive environments encourage users to incorporate more feedback from the environment and remote partners during creative activities. Applicability of findings to the design of new tele-conferencing platforms that incorporate tele-immersive features are further discussed, as well as how they might mitigate some Zoom Fatigue symptoms, but can exacerbate others.

2 METHODOLOGY

Measuring success of improvisation performance is highly subjective. The experimental methodology for previous studies [3] was informed by the insights of Sarker et al. who argue that the

study of virtual teams should begin ‘*inductively, based on the collaborative experiences of virtual team members and the meanings they attribute to the virtual experiences*’ [23]. The results of that research provided the basis for designing survey instruments to empirically support nuanced theories related to how tele-immersion provides greater opportunities for flow, physical engagement, and optimal performance of improvisations between remote partners. We subsequently conducted a cross-disciplinary literature review of linguistics, social psychology, creativity, and improvisational theatre. We used this review to inform a mixed method approach to this follow up study from our previous investigation of improv theatre and tele-immersion [3], in order to account for a complete picture of actual behaviour during the improvisational theatre activity [9, 10, 25, 25], analysing how tele-immersion impacts the collaboration itself.

2.1 Study Design

The collection of audio-visual and survey data in our study was approved by a Research Ethics and Governance Board at the University of Kent. Consent was solicited from participants for use of their data during the study. Data was recorded of participants during remote rehearsals and performance in the co-located virtual space. Fourteen volunteers with varying levels of improvisational experience (1 non-binary, 5 female, 8 male, age between 18-55) were solicited to take part in a 60 minute study of ‘digital tools for improv’. Study participants engaged in three 5-minute improvisational collaborative storytelling sessions with remote partners in the context of a ‘road trip’ using traditional video conferencing interfaces as well as ‘mirror tele-immersed’ interfaces. The loose goal of the improvised scenes was for the participants to engage in a dialogue building towards a creative, collaborative, and cohesive narrative. Participants were introduced to the ‘road trip’ improv format before beginning the study and given 10 minutes to become familiar with the interface and environment, become acquainted with their partner, and ask any questions. Following the induction, the participants were given a randomly selected thematic prompt to inspire a 5 minute ‘road trip scene’. After each 5-minute scene, participants were instructed to immediately fill out a survey of questions related to flow, attunement, presence, the quality of the improvisational experience, and an open ended question where they could describe their general experience. After completing all three ‘drives’ participants were invited to share their impressions of each interface and overall experience. During the open ended interviews, 5 minute ‘sandboxing’ time as well as the 5 minute ‘road trip’ sessions audio and video was recorded for post-study qualitative analysis.

In order to maintain consistency across experiences and minimise interaction and influence by the facilitator, subjects were prompted by a text caption informing them about the upcoming activity. When the subjects started the first study, the prompt said: ‘We will start the road trip soon. Please take a minute to get to know your partner :)’ After 5 minutes of open exploration with the interface, the prompt told them to ‘Get ready for your 5 minute road trip!’ They were then given a prompt they could use or discard to motivate the scene (i.e. ‘Chewing Gum’). At the end of 5 minutes, another prompt appeared: ‘End Scene. Fantastic!’, followed

¹“a state in which people are so involved in an activity that nothing else seems to matter; the experience is so enjoyable that people will continue to do it even at great cost, for the sheer sake of doing it” [19]

²“attunement is a process that includes two parts: (1) the ability to connect with another person’s experience/sensations(i.e., cognitive and affective empathy); and (2) the communication of that connection to the other person” [11]



Figure 2: Participants simulating touch while ‘tele-immersed’.

by ‘Please fill out the survey and return after you have finished’. Participants were assigned a random partner at the start of the study and performed all three scenes with that partner.

Completing a Latin square of combinations would require a cohort of 12 participants (to have 6 pairs). However, as our recruitment campaign managed to attract more participants, we decided to relax that requirement in order to avoid reducing our overall number of participants. This in practice meant that one of the combinations was over represented in our dataset. We did not observe that the specific over-representation of one combination introduced any observable bias in our results.

The three conditions were organised as: *A: Isolated* (with participants webcam feeds appearing adjacent to each other, each with the same background), *B: Tele-immersed* (with the background of both participants’ video replaced with a virtual ‘driving’ environment where they appear seated next to each other as if driving a car) and *C: Single view* (where participants only see their remote partner with the option to toggle a small video mirror on or off). The three layouts were presented in the following orders: (P)1-4: ABC, (P)3-4: ACB, (P)5-6: BCA, (P)7-8: BAC, (P)9-10: CAB, (P)11-12, (P) 13-14 CBA.

2.2 Tele-Immersive Environment

The three tele-immersive environments were built inside of Touchdesigner³, and the VDO.Ninja⁴ WebRTC service was used to connect and to route participant audio and video from their webcams and microphones into the Touchdesigner platform. The webRTC video feed of two participants was fed to background segmentation then superimposed onto a driving video backplate. A simple user interface was built for the facilitator to switch between scenes and direct participants through video text that would appear on their screens. A function was built to record video of each session as well as separate audio from each participant captured directly from their microphone; both audio and video data were stored and catalogued on a private server.

2.3 Surveys

After all three drives had taken place, participants for the given session were invited to fill out a survey and share their general

thoughts and impressions about their experience. The survey questions were adapted from the *Core Flow Scale* designed by Martin and Jackson based on Csikszentmihalyi’s nine dimensions of flow [11, 15], *The Patient’s Experience of Attunement Scale* [26], and improvisational theatre related performance language, and areas of future study recommended by the outcomes of previous studies on tele-immersive theatre [3].

Flow refers to ‘an almost automatic, effortless, yet highly focused state of consciousness’ [11] that would correlate with ‘feeling immersed and performing better improv’. *Attunement* refers to what degree people feel seen and heard by each other in a given social exchange [13].

Questions were presented with a 5 point Likert scale with the following categories: strongly agree, somewhat agree, neutral, somewhat disagree, and strongly disagree. The questions have been formed around the unique experience of performing improvisational theatre online, of which there is no existing validated instrument. Instead an original instrument was formed out of the related instruments for flow, attunement, observations related to physical presence from our previous study, as well as literature related to improvisational studies.

2.4 Dialogue Analysis and Linguistic Measures of Improvisational Technique

Audio from each session (drives and interviews) was transcribed using the Google speech transcription service, reviewed independently by three different members of the research team and subsequently coded qualitatively for emerging patterns. We identified 9 aspects of performances⁵ related to the quality and style of performance that differed across the three platforms. First we noticed a difference in the amount of dialogue in different sessions, particularly as tele-immersive scenes appeared to have more silences and physical interactions than others. Furthermore sessions without a mirror featured more monologues (uninterrupted speech) from individuals than other sessions. We also noticed that some scenes referred more frequently to the given virtual environment, while others did not. In reviewing the footage we also noticed that we could identify when improvisers ‘accepted’ or ‘agreed’ to the offers of their partners and added to them (ie P(1) ‘You kept the mix tape!’/ P(2) ‘Yes, I play it every day’), and when they ‘rejected’ offers

³<https://derivative.ca/>

⁴<https://vdo.ninja/>

⁵These 9 categories refer to various factors known to affect performance and often discussed in improvisational training scenarios.

in the from of suggesting ‘new topics’ (ie $P(3)$ ‘Pepsi is better than Coke’/ $P(4)$ ‘This is a nice view’. The ‘tense’ of the conversations also appeared to change across conditions with many scenes almost exclusively referring to fictional ‘past’ or ‘anticipated’ events, while other scenes appeared to involve more ‘real-time’ problem solving in the moment, i.e. trying to get ‘gum from the glove compartment’ ($p2$). Ultimately, we analysed the semantic choices from the transcripts quantitatively to measure the frequency of each of these 9 features:

1) **Word Count:** Subjective review of transcripts suggested there was more silent ‘physical interaction’ in tele-immersive environments, further suggesting these scenes might have a lower word count than others. However, when word count was calculated, no statistically significant differences in word count were noticed between environments.

2) **Pronoun Usage:** Casual observation of transcripts indicated participants might use more intrinsically motivated language in some scenes, and more extrinsically motivated language in others. To capture the nature of referential language, we tabulated the usage of pronouns into: “I, me, mine, I’ve, I’d”, “You, your, you’re, yours”, and “we, us, our(s)”.

3) **Acceptance of offers:** A standard improvisational trope is that an improviser should ‘accept the offers’ of their partner. To determine the frequency of offer acceptance, transcripts were annotated for how frequently subsequent lines of dialogue incorporated previous statements from a partner.

4) **Adding to offer:** In addition to accepting the offers of scene partners, improvisers are encouraged to ‘add’ (“yes, and”) to the suggestion. We counted the addition of information related to previous offers in each line of dialogue.

5) **Conflict:** Generally improvisers are encouraged to establish a common ground and encourage playful co-creation of narratives. Dialogue was annotated for lines which appeared to introduce a ‘conflict’ related to the previous statement.

We also annotated 6) **references to the visual environment** and to 7) **past**, 8) **present**, and 9) **future events**.

Some features relied on word count and word frequency, which could be counted with Python scripts and Microsoft Excel. Others required manual semantic annotation. To facilitate the annotation, we employed the generative pre-trained transformer (GPT-3) language model [4]. GPT-3 has been shown to have a few-shot learning capability for simple natural language processing tasks such as translation, part-of-speech tagging, etc. and thus we used it to qualify whether each individual line of dialogue was referring to “past”, “present”, “future” events, or whether a pair of consecutive lines of dialogue corresponded to an agreement or disagreement. Note that these preliminary annotations suffer from the lack of reliability of automated NLP methods, hence their results were reviewed and corrected by hand⁶.

⁶The inclusion of language model-based annotation for this study is presented for later discussion on how language models might be used to allow the study of larger improvisational theatre sample sizes.

3 RESULTS

The data results are divided into three categories: *Survey Results* (Section 3.1), *Quantitative Analysis of Improvisational Dialogue* (Section 3.2) and *Qualitative Analysis of Interviews* (Section 3.3). Where appropriate, survey data is compiled and presented alongside observations from the participants about their experience, as well as observations by the researchers of audio and video data from the performances. We performed a Kruskal-Wallis one-way analysis of variance to determine if there was statistical significance⁷ for any of the survey results or any of the 9 features identified in improvisational dialogue (see Section 2.4).

3.1 Survey Results

A rank based non parametric Kruskal-Wallis H test indicated statistically significant differences between tele-immersion and video-feedback in participant self-reporting ($p < 0.05$): “I found myself physically engaging with my whole body”, “I felt physically connected with my partner”, and “I felt transported in the driving scenes”. A near statistical significance ($p < 0.08$) was reported for “I was creatively inspired by the visual environment of the drive”.

The remaining survey results did not yield statistically significant results, however charts on Figure 3 indicate small trends in user experience relevant for discussion and point to areas for further research with larger sample sizes.

3.2 Analysis of Improvisational Dialogue

A rank based non parametric Kruskal-Wallis H test indicated statistically significant ($p < 0.05$) differences between tele-immersive views and video-feedback, in the frequency of the following measures: ‘Present tense’ verb usage and semantic context of the ‘present moment’, and references to the virtual environment during the scene. Charts on Figure 4 indicate that ‘tele-immersion’ was the significantly correlated interface for both categories.

3.3 Interviews

Qualitative analysis of participant feedback after each drive and from post-study interviews supported findings from survey responses. 12/14 participants commented similarly that tele-immersion “was a lot more enjoyable” ($p6$) than the other two interfaces. 9 participants reported variations of the feeling that with tele-immersion “it was easier to physically respond and to see both myself and my partner” ($p1$). 8 indicated taking the most inspiration from the tele-immersed virtual environment, commenting how it was “great being in the same car, and the scenery helped with context and information” ($p12$). The feeling of connection and attunement was most frequently commented on with both the tele-immersed environment (6/14) as well as no-mirror (5/14), but for different reasons. With tele-immersion, participants commented how “being able to play physically... felt more natural and I was a lot more focused on and able to respond to my partner” ($p11$). Without a mirror other participants

⁷The Kruskal-Wallis test ‘is a non-parametric method for testing whether samples originate from the same distribution’. It is used to indicate if ‘at least one sample stochastically dominates one other sample’ [16]. It assumes a null hypothesis that the medians of all groups are equal, and indicates when that hypothesis can be rejected as a result of a significant variance in one of the samples from the others. The results do not indicate which group is significantly different, only that there is one or more samples with a significantly different mean ranking between groups.

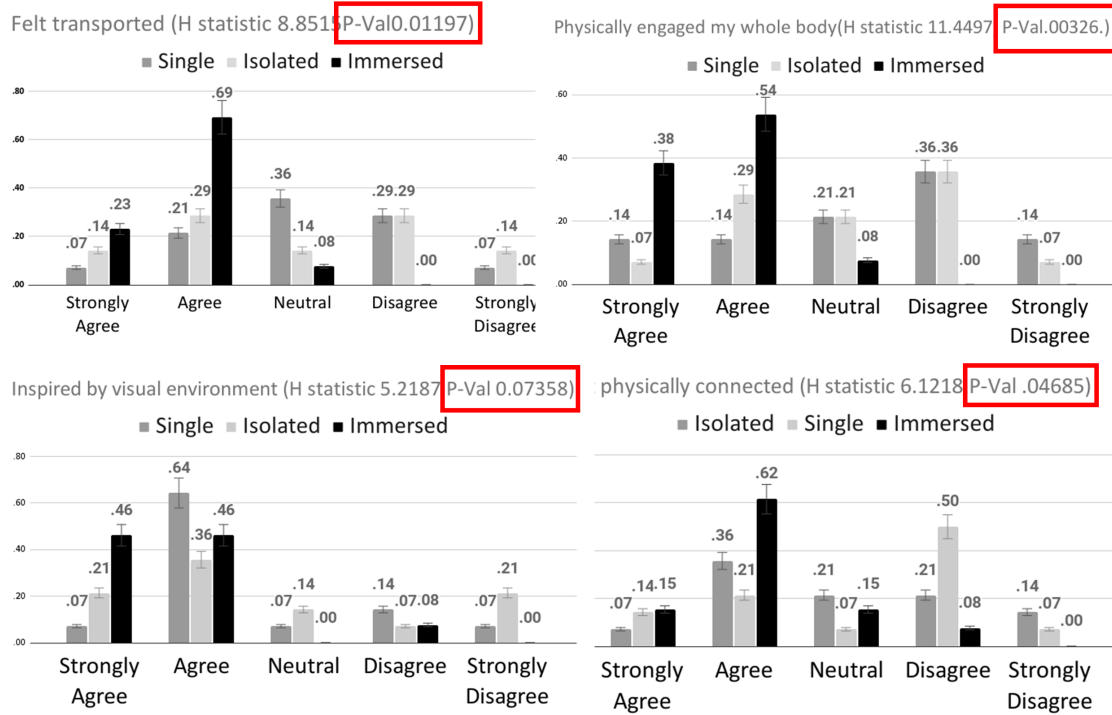


Figure 3: Survey responses with significant ($p < .05$) and near significant ($p < .08$) results. Participants reported physical engagement, sense of physical connection with partner, and feeling transported into the scene as highest with tele-immersion. Taking inspiration from the visual environment was nearly significantly associated with tele-immersion, 2022.

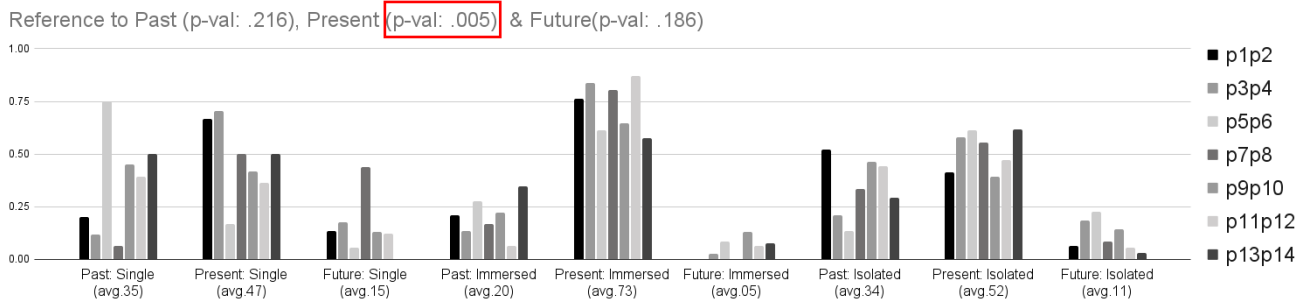


Figure 4: Significant ($p < .05$) results of dialogue analysis. Participants referred to virtual environment and the present moment significantly more often while tele-immersed, 2022.

shared how “not being able to see myself meant I was more focused on what my scene partner was doing” (p1). Few commented directly on their experience of flow, however 6/14 indirectly described how the presence of an isolated mirror negatively affected their focus and how they “felt quite disconnected because I was unable to see my body in the car” (p13). A strong sense of presence with the scene was reported for tele-immersion by 4 participants who described similar sentiments to being “very drawn into the space of the car journey” (p4), and 1 describing how they “felt the car movement” (p3). For 3

participants, not having a mirror however also appeared to be beneficial in fostering a sense of presence, “because you can sometimes find that you check your own appearance instead of focusing on your partner” (p2).

4 DISCUSSION

This study set out to answer whether tele-immersive ‘mirror’ configurations provide advantages over traditional video conferencing platforms to mitigate some symptoms of ‘Zoom Fatigue’ for improvisers, by enhancing the sense of presence between performers, the

experience of flow, the quality of attunement as well as the physical engagement while improvising. Furthermore, we asked if the presence or absence of the video feedback mirror in the different configurations had a measurable impact on improvisational focus and storytelling subject matter and style.

4.1 Designing Tele-Conference Platforms for Performance and Mitigating Symptoms of ‘Zoom Fatigue’

4.1.1 Physical Engagement. While being tele-immersed, participants were able to overcome the inclination to be stationary credited for contributing to ZF associated with traditional conferencing environments. Observing the drives, it appeared that participants were visually motivated to physically engage with the virtual scenery and to simulate naturalistic poses and postures as well as physical contact with their partners. This interaction correlated with a ludic response as participants described the enjoyment and fun they experienced by playfully interacting with each other. Interestingly, the playfulness and physical engagement did not necessarily translate to the sensation of performing better improvisation as some reported that physical engagement came at the cost of being focused more directly on their partner and developing the improvised scene. These observations suggest that tele-immersion while, potentially mitigating some ZF symptoms may not be ideal for all types of remote collaboration, but instead may be best used for specific tasks, with the ability to easily switch to other layouts more appropriate for the given activity. For instance, tele-immersion might be useful for rehearsing scripted theatre before a performance to discover types of physical engagement that can unlock different behaviours, before switching to a no VF interface for rehearsing dialogue. For general remote collaboration, meetings might be structured with short breaks that include tele-immersion as a way to energise participants, and trigger playful engagement before returning to the main collaborative task.

4.1.2 Referencing the Visual Environment and the Now. A significant finding of the study was that during the sessions, participants intuitively made more references to not only the visual environment, but also the dialogue itself stayed in the present moment. Whereas in the other two scenarios, participants started scenes referencing a past event or a character outside of the scene (i.e. comparing who received more Christmas gifts when they were children), the tele-immersed participants tended to generate dialogue around being together in the car in that moment (i.e. searching the car for tarpaulin). For improvisational theatre, this is interesting as it demonstrates tele-immersion is drawing the focus of the improviser externally, which is important for maintaining a more collaborative approach to scene building. These initial findings along with previous observations related to tele-immersive improv [3] point to opportunities for more research into how platforms can be designed specifically for improvisational training that encourages improvisers to learn how to better incorporate external offers into their spontaneous storytelling.

4.1.3 Tele-Immersion for Designers. Beyond improvisational theatre, the finding that tele-immersion encourages an external focus on the visual environment may present opportunities for other

fields or practices that are interested in getting feedback from clients or focus groups about their product, such as interior designers or architects. Some possible applications include designing new social spaces for a building where there is an interest to explore how the colours or shapes of a space or even the artwork on a wall might influence behaviour and how people engage. Potential users of the space could be tele-immersed in the environments and encouraged to have conversations that are later analysed for emergent themes the way the improvisational dialogue was analysed.

4.1.4 Psychosocial Applications of Tele-Immersion. Outside of theatrical applications, this finding has potential relevance for remote collaborations that have a goal of responding in realtime to the behaviour and given environment. Psychosocial experiences might also benefit from exploring this feature of tele-immersion, where it is important that participants stay in the present moment. Couples therapy, conflict resolution, or public debates might each find interesting applications of a tele-immersive environment that encourages participants to stay focused on what is happening right now, rather than focusing on past or future events.

4.2 Limitations and Future Work

Given the small sample size, it was interesting that we observed significant results in many areas. However these observations should not be used to make too general an argument about the effects of tele-immersion, since participants only engaged with the environments for relatively short periods. Instead, the findings provide support for future research into the features of tele-immersion for larger and more diverse groups. Moreover, these results add some empirical weight to previous anecdotal reports on the power of tele-immersion to make people feel like they are really ‘there’ in the virtual world. The findings further provide an impetus for conducting longer studies of people working in tele-immersed environments, as well as other possible configurations to examine how social interactions and dialogue might be affected. It is also important to note, that users relied on their own internet speeds to connect, and there was variable quality of connection and experience of ‘lag’ during the trials. In future studies, we plan to control for variability in hardware and connection speed by performing tests in a laboratory setting, as well as comparing tele-immersion experienced on a screen to tele-immersion experienced with a virtual reality headset.

5 CONCLUSION

In this paper we compared the effects of video feedback (VF) presented as an isolated window, within a tele-immersive environment, and no video feedback at all, on a remote collaboration task. Our novel field study of 14 domain experts performing improvisational theatre under all three conditions produced significant results from our mixed method study. Results from surveys, participant interviews, and quantitative analytics of the dialogue produced during improvisational scenes, found that tele-immersion significantly impacted five different domains of performance. ‘Physical engagement’ and ‘presence’ were perceived highest with tele-immersion, while ‘attunement’ and ‘flow’ between performers were comparable between tele-immersion and no mirror. Isolated VF was perceived

worse for most conditions. Analysis of improvisation dialogue indicated TI significantly impacted verb tense and direct references to a given scenic environment.

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Appendices

A APPENDIX

A.1 Survey Questions

- I was 'totally involved'.
- I was 'tuned in' to what I was doing.
- I was 'in the zone'.
- I felt completely immersed in the activities.
- I enjoyed performing with my scene partners.
- My driving companion seemed to enjoy working with me today.
- My driving companion's comments enabled me to be creative today.
- My driving companion understood me.
- What my driving companion did and said was helpful today
- My driving companion seemed comfortable with all of my reactions today
- I felt accepted by my driving companion today
- I felt a sense of having a meaningful person-to-person relationship with my driving companion today
- I felt fully present with my partner.
- I found myself physically engaging with my body.
- I was very aware of my partner's body.
- I was often making extra effort to think of interesting things to say.
- I become so immersed in the scene that I forget my physical surroundings.
- I was distracted by my physical surroundings.
- It was easy for me to add to and participate in the conversation.
- I was often making effort to be funny.
- I relied significantly on my partner's reactions to inform my performance.
- I was very aware of my partners reactions.
- I felt physically connected with my partner.



Supplementary Figure 1: Screenshot of TouchDesigner network developed for the study and the GUI for controlling drives, 2022.

- I was inspired creatively by the visual environment of the drive.
- I was distracted by the visual environment of the drive.
- I felt transported into the driving scene.
- My ability to feel connected was limited by the technology used.

- Please describe your general experience of DriveTogether during this most recent session.

A.2 Tele-Immersive Environment Design

See Supplementary Figure 1.